

EXPERIMENT-6

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Subject-Advance DBMS

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Section-24AIT KRG 1_B

(A) Problem Statement

A fruit distribution company records the number of apples and oranges sold each day. For every date, exactly one record exists for apples and one record exists for oranges.

Generate a daily sales comparison report showing the difference between the number of apples sold and the number of oranges sold on the same day. The difference should be calculated as:

Difference = Apples Sold - Oranges Sold

Display the sale date and the calculated difference. The result should be sorted by sale date in ascending order.

```
SELECT
  sale_date,
  SUM(
    CASE
      WHEN fruit = 'apples'
      THEN sold_num
      ELSE -sold_num
    END
  ) AS diff
FROM Sales
GROUP BY sale_date
ORDER BY sale_date;
```

(B) Problem Statement

A company stores hierarchical employee reporting information in a table. Each record represents a node in a hierarchy where:

- id represents the current node.
- p_id represents the parent node.

Classify every node into one of the following categories:

- **Root:** A node that does not have any parent.
- **Leaf:** A node that has a parent but does not have any children.
- **Inner:** A node that has both a parent and at least one child.

Display the node ID along with its category. Return the result sorted by node ID.

```
SELECT
  id,
  CASE
    WHEN p_id IS NULL THEN 'Root'
    WHEN id IN (SELECT p_id FROM tree) THEN 'Inner'
    ELSE 'Leaf'
  END AS type
FROM Tree;
```